

Sushmita Paul

+880-188-170-1129 | sushmitap247@gmail.com
LinkedIn | GitHub | Portfolio

RESEARCH INTERESTS

SLM Reasoning & Evaluation, Reliable AI, Efficient ML Systems, Computer Vision

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering (Honors)

Feb 2020 – Mar 2025

Dhaka, Bangladesh

- **CGPA:** 3.80/4.00 | **Award:** Dean's List, BUET
- **Selected Coursework:** Artificial Intelligence, Machine Learning, Operating Systems, Data Structures & Algorithms.

RESEARCH EXPERIENCE

External State Support and History Management in Small Language Models for Long-Horizon Graph Traversal

May 2025 – Present

Research Collaboration with Prof. Ch. Md. Rakin Haider

BUET

- Investigating **state tracking and reasoning reliability** in small language models (SLMs) through multi-turn breadth-first and depth-first graph traversal, with controlled comparisons across Qwen and Gemma models, task difficulty, and history-retention policies.
- Implemented four state-support conditions: no injection, prose reminders, structured state registries, and **Verified after Injection (VI)**. VI checks model-reported state against the environment's ground truth and supplies targeted corrections for missing, malformed, or inaccurate reports.
- Built a reproducible **vLLM/Slurm evaluation pipeline** with fixed data splits, frozen experiment manifests, resumable runs, and trajectory logs; measures task completion, traversal correctness, invalid actions, and recovery after errors.

Detection of Dengue Breeding Sites in Unplanned Urban Landscapes with Deep Learning from Remote Sensing

Nov 2023 – Jan 2025

Undergraduate Thesis with Prof. Anindya Iqbal [Thesis]

BUET

- Developed a **two-stage framework** combining breeding-object detection and building segmentation through spatial containment analysis; achieved **83.6% balanced accuracy**, a 14.1 percentage-point gain over a joint detection-classification baseline.
- Derived a combinatorial detection analysis showing identification probability above 0.8 with at least four rooftop objects under the modeled object-type distributions, despite individual class mAP50 below 0.36.
- Created a UAV imagery dataset with **5,845 objects across six classes and 289 building boundaries**; evaluated YOLOv9 GELAN-C, SegGPT, YOLOv8m, and YOLOv11m.
- Designed sliding-window postprocessing with modified Non-Maximum Merging to reconstruct buildings across overlapping image slices; validated 50% overlap through sensitivity analysis.
- Designed a spatially informed inspection strategy achieving 89.1% true-positive and 78.1% true-negative rates, with an estimated **35% reduction in inspection labor costs**.

MANUSCRIPTS & PRESENTATIONS

1. **Paul, S.**, et al. “Detection of Dengue Breeding Sites in Unplanned Urban Landscapes with Deep Learning from Remote Sensing.” Manuscript under review at *BMC Public Health* (Springer Nature). *Co-first author*.
2. **Paul, S.**, et al. “Identification of Dengue Breeding Sites from Point-Cloud Images.” *BEAR Summit 2025*, Bangladesh. Digital poster presentation.

SELECTED RESEARCH PROJECT

Medical Image Analysis: Lung Nodule Detection & Segmentation

Jan 2025

Machine Learning Course Project [[Code](#)]

BUET

- Evaluated SAM2 and SwinUNETR for lung nodule segmentation and Faster R-CNN for detection on LIDC-IDRI; used hybrid Dice/BCE loss, mixed-precision training, and gradient accumulation.
- Achieved **Dice 0.93, precision 0.96, and recall 0.89**; reconciled multi-annotator masks using consensus-based ground truth.

PROFESSIONAL EXPERIENCE

Therap BD Ltd

Apr 2025 – Present

Machine Learning Engineer

Dhaka, Bangladesh

- Developed LLM pipelines for clinical text summarization, including healthcare data extraction, prompt design, hallucination mitigation, multi-model evaluation, and adversarial robustness testing.
- Benchmarked vision-language and segmentation models for deployment; built a real-time face detection system and optimized NVIDIA Jetson inference using TensorRT quantization.
- Compared encoder and decoder architectures for text classification, examining model accuracy and deployment trade-offs.

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, SQL

ML Frameworks & Methods: PyTorch, scikit-learn, Hugging Face, ONNX, Optuna, GANs, VAEs, diffusion models

Language Models: LoRA/QLoRA fine-tuning, vLLM, reasoning evaluation, prompt engineering, vision-language models, ModernBERT

Computer Vision: SAM/SAM2, YOLO, SwinUNETR, Faster R-CNN, OCR

Systems: Slurm, Linux, Git, Docker, Azure, NVIDIA Jetson, TensorRT, FP16/INT8 quantization, Celery, RabbitMQ

TEACHING & OUTREACH

- Tutored Higher Secondary Certificate students in Mathematics and Physics; delivered a technical seminar at Islamic University of Technology (IUT).